Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A particle moving with uniform retardation along a straight line covers distances a and b in successive intervals p and q seconds. The acceleration of the particle is

(A) $\frac{2(aq-bp)}{pq(p+q)}$

(B) $\frac{2(bp-aq)}{pq(p+q)}$

(C) $\frac{2(aq+bp)}{pq(p+q)}$

(D) $\frac{2(aq-bp)}{pq(p-q)}$

Q.2

A goods train accelerating uniformly on a straight railway track, approaches an electric pole standing on the side of track. Its engine passes the pole with velocity u and the guard's room passes with velocity v. The middle wagon of the train passes the pole with a velocity

(A) $\frac{u+v}{2}$

(B) $\frac{1}{2}\sqrt{u^2+v^2}$

(C) $\sqrt{uv}$

(D) $\sqrt{\left(\frac{u^2+v^2}{2}\right)}$

Q.3

A car, starting from rest, accelerates at the rate f through a distance S, then continues at constant speed for time t and then decelerates at the rate $\frac{f}{2}$ to come to rest. If the total distance traversed is $15S$, then

(A) $S=\frac{1}{6}ft^2$

(B) $S=ft$

(C) $S=\frac{1}{4}ft^2$

(D) $S=\frac{1}{72}ft^2$

Q.4

A stone falls freely under gravity. It covers distances h_1, h_2 and h_3 in the first 5 seconds, the next 5 seconds and the next 5 seconds respectively. The relation between $\mathrm{h}_1,\ \mathrm{h}_2$ and h_3 is

(A) $\mathrm{h}_1=\frac{\mathrm{h}_2}{3}=\frac{\mathrm{h}_3}{5}$

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(B) $\mathrm{h_2 = 3\ h_1}$ and $\mathrm{h_3 = 3\ h_2}$

(C) $\mathrm{h_1 = h_2 = h_3}$

(D) $h_1 = 2h_2 = 3h_3$

Q.5

A particle starts its motion from rest under the action of a constant force. If the distance covered in first 10 seconds is S_1 and that covered in the first 20 seconds is S_2, then:

(A) $\mathrm{S_2 = 3\ S_1}$

(B) $\mathrm{S_2 = 4\ S_1}$

(C) $\mathrm{S_2 = S_1}$

(D) $\mathrm{S_2 = 2\ S_1}$

Q.6

A body moving with a uniform acceleration crosses a distance of $65\ \mathrm{m}$ in the 5 th second and $105\ \mathrm{m}$ in 9 th second. How far will it go in 20 s?

(A) $2040\ \mathrm{m}$

(B) $240\ \mathrm{m}$

(C) $2400\ \mathrm{m}$

(D) $2004\ \mathrm{m}$

Q.7

A bullet fired into a wooden block loses half of its velocity after penetrating $40\ \mathrm{cm}$. It comes to rest after penetrating a further distance of

(A) $\frac{22}{3}\ \mathrm{cm}$

(B) $\frac{40}{3}\ \mathrm{cm}$

(C) $\frac{20}{3}\ \mathrm{cm}$

(D) $\frac{22}{5}\ \mathrm{cm}$

Q.8

An automobile travelling with a speed of $60\ \mathrm{km/h}$, can brake to stop within a distance of $20\ \mathrm{m}$. If the car is going twice as fast i.e., $120\ \mathrm{km/h}$, the stopping distance will be

(A) $60\ \mathrm{m}$

(B) $40\ \mathrm{m}$

(C) $20\ \mathrm{m}$

(D) $80\ \mathrm{m}$

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Q.9

The water drops fall at regular intervals from a tap $5\ \mathrm{m}$ above the ground. The third drop is leaving the tap at an instant when the first drop touches the ground. How far above the ground is the second drop at that instant? (Take $g = 10\ \mathrm{m/s^2}$)

(A) $1.25\ \mathrm{m}$

(B) $2.50\ \mathrm{m}$

(C) $3.75\ \mathrm{m}$

(D) $5.00\ \mathrm{m}$

Q.10

A metro train starts from rest and in five seconds achieves a speed $108\ \mathrm{km/h}$. After that it moves with constant velocity and comes to rest after travelling $45\ \mathrm{m}$ with uniform retardation. If total distance travelled is $395\ \mathrm{m}$, find total time (in sec) of travelling.

(A) 17.5

(B) 17.6

(C) 17.1

(D) 17.2

Q.11

A ball is dropped from the top of a tower of height $100\ \mathrm{m}$ and at the same time another ball is projected vertically upwards from ground with a velocity $25\ \mathrm{ms^{-1}}$. Then what will be the distance (in m) from the top of the tower, at which the two balls meet?

(A) 78.9

(B) 78

(C) 78.1

(D) 78.4

Q.12

A car, moving with a speed of $50\ \mathrm{km/hr}$, can be stopped by brakes after at least $6\ \mathrm{m}$. If the same car is moving at a speed of $100\ \mathrm{km/hr}$, then what is the minimum stopping distance (in m)?

(A) 26

(B) 27

(C) 23

(D) 24

Q.13

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The distances travelled by a body starting from rest and travelling with uniform acceleration in successive intervals of time each of one second will be in the ratio:

(A) $1:2:3$

(B) $1:2:4$

(C) $1:3:5$

(D) $1:5:9$

Q.14

A bus starts moving with acceleration $2\ \mathrm{m/s^2}$. A cyclist $96\ \mathrm{m}$ behind the bus starts simultaneously towards the bus at $20\ \mathrm{m/s}$. After what time will he be able to overtake the bus?

(A) $4\mathrm{sec}$

(B) $8\mathrm{sec}$

(C) $18\mathrm{sec}$

(D) $16\mathrm{sec}$

Q.15

A body covers $26, 28, 30, 32$ meters in 10th , 11th , 12th and 13th seconds respectively. The body starts

(A) from rest and moves with uniform velocity

(B) from rest and moves with uniform acceleration

(C) with an initial velocity and moves with uniform acceleration

(D) with an initial velocity and moves with uniform velocity

Q.16

A particle crossing the origin of co-ordinates at time $\mathrm{t} = 0$, moves in the xy -plane with a constant acceleration a in the y-direction. If its equation of motion is $y = bx^2$ (b is a constant), its velocity component in the x -direction is

(A) $\sqrt{\frac{2\,\mathrm{b}}{\mathrm{a}}}$

(B) $\sqrt{\frac{a}{2b}}$

(C) $\sqrt{\frac{a}{b}}$

(D) $\sqrt{\frac{b}{a}}$

Q.17

The distance time graph of a particle at time makes angles 45° with the time axis. After one second, it makes angle 60° with the time axis. What is the acceleration of the particle?

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(A) $\sqrt{3}-1$

(B) $\sqrt{3}+1$

(C) $\sqrt{3}$

(D) 1

Q.18

A particle experiences constant acceleration for 20 seconds after starting from rest. If it travels a distance s_1 in the first 10 seconds and distance s_2 in the next 10 seconds, then

(A) $s_2 = s_1$

(B) $s_2 = 2\ s_1$

(C) $s_2 = 3\ s_1$

(D) $s_2 = 4\ s_1$

Q.19

A particle moves along the curve $\frac{x^2}{9}+\frac{y^2}{4}=1$, with constant speed v. Express its velocity vectorially as a function of (x, y).

(A) $\frac{(\mp 9y\hat{i} \pm 4x\hat{j})v}{\sqrt{16x^2+81y^2}}$

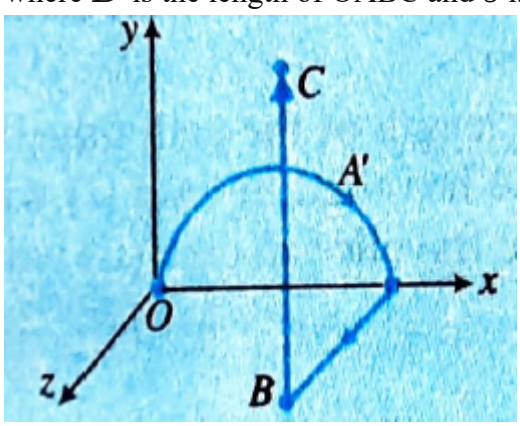
(B) $\frac{(\mp 9y\hat{i} \pm 7x\hat{j})v}{\sqrt{16x^2+81y^2}}$

(C) $\frac{(\mp 9y\hat{i} \pm 9x\hat{j})v}{\sqrt{16x^2+81y^2}}$

(D) $\frac{(\mp 9y\hat{i} \pm 5x\hat{j})v}{\sqrt{16x^2+81y^2}}$

Q.20

A particle moves in a semicircular path of radius R from O to A. Then it moves parallel to z-axis covering a distance R upto B. Finally it moves along BC parallel to y-axis through a distance $2R$. Find the ratio of D/s, where D is the length of OABC and s is the magnitude of $\vec{s}$ shown in the diagram.



(A) $\frac{\pi+3}{3}$

(B) $\frac{\pi+4}{4}$

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(C) $\frac{\pi+5}{5}$

(D) $\frac{\pi+6}{6}$

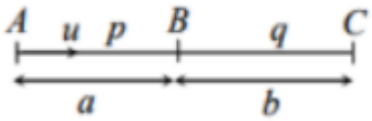
Answers & Solutions

Q.1 Answer:

$\frac{2(aq-bp)}{pq(p+q)}$

Solution:

Let retardation be f and initial velocity be u. For AB,



Let retardation be f and initial velocity be u.

For AB

$$a = up - \frac{1}{2}p^2 \quad ...(i)$$

$$\frac{a}{p} = u - \frac{fp}{2}$$

For AC,

$a + b = u(p+q) - \frac{1}{2}f(p+q)^2$ **...(ii)**

$\frac{a+b}{p+q} = u - \frac{f(p+q)}{2}$

Subtracting (i) and (ii)

$$\frac{a+b}{p+q} - \frac{a}{p} = \frac{fp}{2} - \frac{f(p+q)}{2}$$

$$\frac{pa+pb-ap-aq}{p(p+q)} = \frac{fq}{2}$$

$$f = \frac{2(aq-bp)}{pq(p+q)}$$

Q.2 Answer:

$\sqrt{\left(\frac{u^2+v^2}{2}\right)}$

Solution:

Let ' S' be the distance between two ends and ' a ' be the constant acceleration As we know $v^2 - u^2 = 2aS$

or, $aS = \frac{v^2-u^2}{2}$

Let v_c be velocity at mid point. Therefore, $v_c^2 - u^2 = 2a\frac{S}{2}$

$v_c^2 = u^2 + aS$

$v_c^2 = u^2 + \frac{v^2-u^2}{2}$

$v_c = \sqrt{\frac{u^2+v^2}{2}}$

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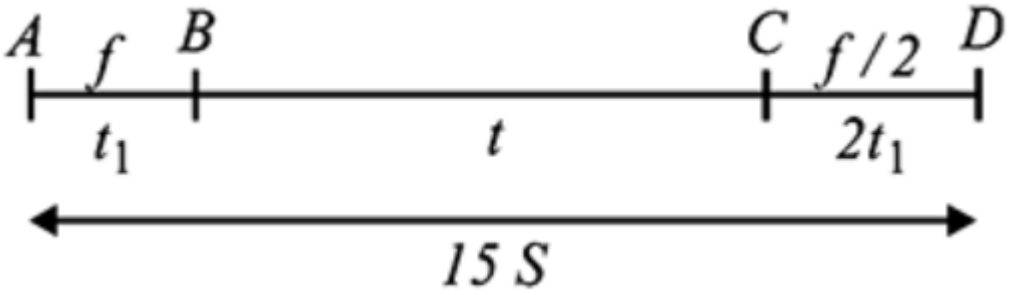
Q.3 Answer:

$S = \frac{1}{72} f t^2$

Solution:

Distance from A to $B = S = \frac{1}{2} f t_1^2$ Distance from B to $C = (f t_1)\, t$

Distance from C to $D = \frac{u^2}{2a} = \frac{(f t_1)^2}{2(f/2)} = f t_1^2 = 2S$



$\Rightarrow \quad S + f t_1 t + 2S = 15S$

$\Rightarrow \quad f t_1 t = 12S$

..... (i)

$\frac{1}{2} f t_1^2 = S$

Dividing (i) by (ii), we get $t_1 = \frac{t}{6} \Rightarrow S = \frac{1}{2} f \left(\frac{t}{6}\right)^2 = \frac{f t^2}{72}$

Q.4 Answer:

$h_1 = \frac{h_2}{3} = \frac{h_3}{5}$

Solution:

$\because h = \frac{1}{2} g t^2$

$\therefore h_1 = \frac{1}{2}\, g(5)^2 = 125$

$h_1 + h_2 = \frac{1}{2}\, g(10)^2 = 500$

$\Rightarrow h_2 = 375$

$h_1 + h_2 + h_3 = \frac{1}{2}\, g(15)^2 = 1125$

$\Rightarrow h_3 = 625$

$h_2 = 3\, h_1,\ h_3 = 5\, h_1$

or $h_1 = \frac{h_2}{3} = \frac{h_3}{5}$

Q.5 Answer:

$S_2 = 4\, S_1$

Solution:

$u = 0, t_1 = 10\ s, t_2 = 20\ s$

Using the relation, $S = ut + \frac{1}{2} a t^2$

Acceleration being the same in two cases, $S_1 = \frac{1}{2} a \times t_1^2,\ S_2 = \frac{1}{2} a \times t_2^2$

$\therefore \quad \frac{S_1}{S_2} = \left(\frac{t_1}{t_2}\right)^2 = \left(\frac{10}{20}\right)^2 = \frac{1}{4}$

$S_2 = 4\, S_1$

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Q.6 Answer:

$2400\ \mathrm{m}$

Solution:

We have, $S_n = u + \frac{a}{2}(2n-1)$
or $65 = \mathrm{u} + \frac{a}{2}(2 \times 5 - 1)$
or $65 = \mathrm{u} + \frac{9}{2}a$
Also, $105 = \mathrm{u} + \frac{a}{2}(2 \times 9 - 1)$
or $105 = \mathrm{u} + \frac{17}{2}a$
Equation $(2) - (1)$ **gives,**
$40 = \frac{17}{2}a - \frac{9}{2}a = 4a$ **or** $a = 10\ \mathrm{m/s^2}$
Substitute this value in (1) we get, $\mathrm{u} = 65 - \frac{9}{2} \times 10 = 65 - 45 = 20\ \mathrm{m/s}$
∴ The distance travelled by the body in $20\ \mathrm{s}$ **is,**
$\mathrm{s} = \mathrm{ut} + \frac{1}{2}at^2 = 20 \times 20 + \frac{1}{2} \times 10 \times (20)^2$
$= 400 + 2000 = 2400\ \mathrm{m}$

Q.7 Answer:

$\frac{40}{3}\ \mathrm{cm}$

Solution:

For first part of penetration, by equation of motion $\left(\frac{\mathrm{u}}{2}\right)^2 - (\mathrm{u})^2 = 2\mathrm{aS}$ **or** $\mathrm{a} = -\frac{3\mathrm{u}^2}{8\,\mathrm{S}} \quad \ldots(\mathrm{i})$
For latter part of penetration $(0)^2 - \left(\frac{\mathrm{u}}{2}\right)^2 = 2\mathrm{aS}', \mathrm{S}' = -\frac{\mathrm{u}^2}{8\mathrm{a}}$
$\mathrm{S}' = -\frac{\mathrm{u}^2}{8}\left(\frac{8\,\mathrm{S}}{-3\mathrm{u}^2}\right)$ **(Using (i))**
$\mathrm{S}' = \frac{\mathrm{S}}{3}$ **or** $\mathrm{S}' = \frac{40}{3}\ \mathrm{cm}$

Q.8 Answer:

$80\ \mathrm{m}$

Solution:

Speed, $u = 60 \times \frac{5}{18}\ \mathrm{m/s} = \frac{50}{3}\ \mathrm{m/s}$
$d = 20\ \mathrm{m}, u' = 120 \times \frac{5}{18} = \frac{100}{3}\ \mathrm{m/s}$
Let declaration be a then $(0)^2 - u^2 = -2ad$ **or** $u^2 = 2ad$ **...(1) and** $(0)^2 - u'^2 = -2ad'$
or $u'^2 = 2ad'$
(2) divided by (1) gives,
$4 = \frac{d'}{d} \Rightarrow d' = 4 \times 20 = 80\ \mathrm{m}$

Q.9 Answer:

$3.75\ \mathrm{m}$

Solution:

Height of tap $= 5\ \mathrm{m}$ **and** $(\mathrm{g}) = 10\ \mathrm{m/sec^2}$**. For the first drop,**
$5 = ut + \frac{1}{2}gt^2 = (0 \times t) + \frac{1}{2} \times 10t^2 = 5t^2$ **or** $t^2 = 1$ **or** $t = 1$
It means that the third drop leaves after one second of the first drop. Or, each drop leaves after every $0.5\mathrm{sec}$**. Distance covered by the second drop in** $0.5\mathrm{sec}$

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$= \mathrm{ut} + \frac{1}{2}\mathrm{gt}^2 = (0 \times 0.5) + \frac{1}{2} \times 10 = (0.5)^2 = 1.25\ \mathrm{m}$

Therefore, distance of the second drop above the ground $= 5 - 1.25 = 3.75\ \mathrm{m}.$

Q.10 Answer:

17.2

Solution:

Ist part: $\mathrm{u} = 0, \mathrm{t} = 5\ \mathrm{s}, \mathrm{v} = 108\ \mathrm{km/hr} = 30\ \mathrm{m/s}$

$\mathrm{v} = \mathrm{u} + \mathrm{at} \Rightarrow 30 = 0 + \mathrm{a} \times 5 \Rightarrow \mathrm{a} = 6\ \mathrm{m/s}^2$

$\mathrm{s} = \mathrm{ut} + \frac{1}{2}\mathrm{at}^2 = 0 \times 5 + \frac{1}{2} \times 6 \times 5^2 = 75\ \mathrm{m}$

IIIrd part: $\mathrm{s} = 45\ \mathrm{m}, \mathrm{u} = 30\ \mathrm{m/s}, \mathrm{v} = 0$

$a = \frac{v^2 - u^2}{2s} = \frac{-30 \times 30}{2 \times 45} = -10\ \mathrm{m/s}^2$

$\mathrm{v} = \mathrm{u} + \mathrm{at} \Rightarrow 0 = 30 - 10 \times \mathrm{t} \Rightarrow \mathrm{t} = 3\ \mathrm{s}$

IInd part :

$\mathrm{s} = \mathrm{s}_1 + \mathrm{s}_2 + \mathrm{s}_3$

$395 = 75 + \mathrm{s}_2 + 45 \Rightarrow \mathrm{s}_2 = 275\ \mathrm{m}$

$t = \frac{275}{30} = 9.16 = 9.2\ \mathrm{s}$

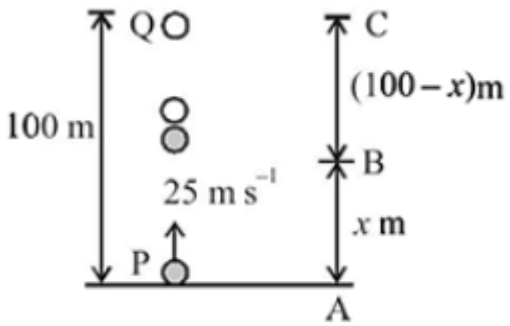
Total time taken $= (5 + 9.2 + 3)\mathrm{sec} = 17.2\mathrm{sec}$

Q.11 Answer:

78.4

Solution:

Let the two balls Pand Q **meet at height** x m **from the ground after time** t s **from the start. We have to find distance,** $BC = (100 - x)$



For ball P $S = x\ \mathrm{m}, u = 25\ \mathrm{m\ s}^{-1}, a = -g$

From $S = ut + \frac{1}{2}at^2$

$x = 25t - \frac{1}{2}gt^2$ .....**(i)**

For ball Q $S = (100 - x)\mathrm{m}, u = 0, a = g$

$\therefore 100 - x = 0 + \frac{1}{2}gt^2$ ..... **(ii)**

Adding eqns. (i) and (ii), we get $100 = 25t$ **or** $t = 4$ s

From eqn. (i), $x = 25 \times 4 - \frac{1}{2} \times 9.8 \times (4)^2 = 21.6\ \mathrm{m}$

Hence distance from the top of the tower $= (100 - x)\mathrm{m} = (100 - 21.6\ \mathrm{m}) = 78.4\ \mathrm{m}$

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Q.12 Answer:

24

Solution:

Case-1: $u = 50 \times \frac{5}{18}\ \mathrm{m/s}$

$v = 0,\ \mathrm{s} = 6\ \mathrm{m}, a = a$

$v^2 - u^2 = 2as$

$\Rightarrow 0^2 - \left(50 \times \frac{5}{18}\right)^2 = 2 \times a \times 6$

$\Rightarrow -\left(50 \times \frac{5}{18}\right)^2 = 2 \times a \times 6 \ldots .(\mathrm{i})$

Case-2 : $u = 100\ \mathrm{km/hr} = 100 \times \frac{5}{18}\ \mathrm{m/sec}$

$v = 0, s = s, a = a \quad \therefore v^2 - u^2 = 2as$

$\Rightarrow 0^2 - \left(100 \times \frac{5}{18}\right)^2 = 2\mathrm{as}$

$\Rightarrow -\left(100 \times \frac{5}{18}\right)^2 = 2as \quad \ldots$

Dividing (i) and (ii) we get $\frac{100 \times 100}{50 \times 50} = \frac{2 \times a \times s}{2 \times a \times 6} \Rightarrow \mathrm{s} = 24\ \mathrm{m}$

Q.13 Answer:

$1 : 3 : 5$

Solution:

Distance traveled (s) is given by:

$\mathrm{s} = \mathrm{ut} + \frac{1}{2}\mathrm{at}^2$

Now, interval wise distance traveled are:

$\mathrm{s}_1 - \mathrm{s}_0 = \frac{\mathrm{a}}{2}(1)^2 = \frac{\mathrm{a}}{2}$

$s_2 - s_1 = \frac{a}{2}(2)^2 - \frac{a}{2}(1)^2 = \frac{3a}{2}$

$\mathrm{s}_3 - \mathrm{s}_2 = \frac{\mathrm{a}}{2}(3)^2 - \frac{\mathrm{a}}{2}(2)^2 = \frac{5\mathrm{a}}{2}$

Hence, its in the ratio $= \frac{\mathrm{a}}{2} : \frac{3\mathrm{a}}{2} : \frac{5\mathrm{a}}{2} = 1 : 3 : 5$

Q.14 Answer:

8sec

Solution:

Velocity of bus $\mathrm{V_b} = 0$, **Velocity of cyclist** $\mathrm{V}_c = 20\ \mathrm{m/s}$,

Acceleration of bus $\mathrm{a_b} = 2\ \mathrm{m/s^2}$ **Acceleration of cyclist** $\mathrm{a_c} = 0$

Relative velocity $\mathrm{V_{cb}} = \mathrm{V_c} - \mathrm{V_b} = 20\ \mathrm{m/s}$

Relative acceleration $\mathrm{a}_{cb} = \mathrm{a_c} - \mathrm{a_b} = -2\ \mathrm{m/s^2}$

Relative separation $= 96\ \mathrm{m}$

Using $s = ut + \frac{1}{2}at^2 \Rightarrow 96 = 20t - \frac{1}{2}2t^2 \Rightarrow t^2 - 20t + 96 = 0$

Solving the equation, we get $t = 8$ **s and** $t = 12$ **s**

Hence, after 8 s, cyclist will overtake the bus.

Q.15 Answer:

with an initial velocity and moves with uniform acceleration

Solution:

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The distance covered in n^{th} second is $S_n = u + \frac{1}{2}(2n-1)a$
where u is initial velocity & a is acceleration then
$26 = \mathrm{u} + \frac{19\mathrm{a}}{2} \quad \ldots(i)$
$28 = \mathrm{u} + \frac{21\mathrm{a}}{2} \quad \ldots(ii)$
$30 = \mathrm{u} + \frac{23\mathrm{a}}{2} \quad \ldots(iii)$
$32 = \mathrm{u} + \frac{25\mathrm{a}}{2} \quad \ldots(iv)$
From eqs. (i) and (ii) we get $\mathrm{u} = 7\ \mathrm{m/sec}, \mathrm{a} = 2\ \mathrm{m/sec}^2$
$\therefore$ The body starts with initial velocity $\mathrm{u} = 7\ \mathrm{m/sec}$ and moves with uniform acceleration $\mathrm{a} = 2\ \mathrm{m/sec}^2$

Q.16 Answer:

$\sqrt{\frac{a}{2b}}$

Solution:

$\mathrm{y} = \mathrm{bx}^2$
Differentiating w.r.t to t an both sides, we get $\frac{dy}{dx} = b2x\frac{dx}{dt}$
$\mathrm{v_y} = 2\mathrm{bxv_x}$
Again differentiating w.r.t to t on both sides we get $\frac{\mathrm{dv_y}}{\mathrm{dt}} = 2\mathrm{bv_x}\frac{\mathrm{dx}}{\mathrm{dt}} + 2\mathrm{bx}\frac{\mathrm{dv_x}}{\mathrm{dt}} = 2\mathrm{bv_x^2} + 0$
$\left[\frac{\mathrm{dv_x}}{\mathrm{dt}} = 0\right.$, because the particle has constant acceleration along y-direction] Now, $\frac{\mathrm{dv_y}}{\mathrm{dt}} = \mathrm{a} = 2\mathrm{bv_x^2}$
$v_x^2 = \frac{a}{2b}$
$v_x = \sqrt{\frac{a}{2b}}$

Q.17 Answer:

$\sqrt{3} - 1$

Solution:

Velocity at time t is $\tan 45^\circ = 1$. Velocity at time $(t = 1)$ is $\tan 60^\circ = \sqrt{3}$. Acceleration is change in velocity in one second $= \sqrt{3} - 1$.

Q.18 Answer:

$\mathrm{s_2} = 3\ \mathrm{s_1}$

Solution:

Let a be the constant acceleration of the particle. Then $\mathrm{s} = \mathrm{ut} + \frac{1}{2}\mathrm{at}^2$
$\mathrm{s_1} = 0 + \frac{1}{2} \times \mathrm{a} \times (10)^2 = 50$
and $\mathrm{s_2} = \left[0 + \frac{1}{2}\mathrm{a}(20)^2\right] - 50\mathrm{a} = 150\mathrm{a}$
$\therefore \mathrm{s_2} = 3\ \mathrm{s_1}$

Q.19 Answer:

$\frac{(\mp 9y\hat{i} \pm 4x\hat{j})v}{\sqrt{16x^2+81y^2}}$

Solution:

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Given $\frac{x^2}{9}+\frac{y^2}{4}=1$

Differentiating both sides w.r.t. time $\Rightarrow \quad \frac{1}{9}\frac{dx^2}{dt}+\frac{1}{4}\frac{dy^2}{dt}=\frac{d(1)}{dt}$

$\Rightarrow \quad \frac{1}{9}2x\cdot\frac{dx}{dt}+\frac{1}{4}\cdot 2y\cdot\frac{dy}{dt}=0$

$\Rightarrow \quad \frac{2x}{9}\left[\frac{dx}{dt}\right]+\frac{2y}{4}\left[\frac{dy}{dt}\right]=0$

$\Rightarrow \quad v_x=-\frac{9y}{4x}v_y \quad \left[\frac{dx}{dt}=v_x;\frac{dy}{dt}=v_y\right]\ldots(\mathrm{i})$

As particle moves with constant velocity, then

$v_x^2+v_y^2=v^2\Rightarrow\left(-\frac{9}{4}\frac{y}{x}v_y\right)^2+v_y^2=v^2$

$\Rightarrow \quad v_y^2=\frac{16x^2v^2}{16x^2+81y^2}\Rightarrow v_y=\frac{\pm 4xv}{\sqrt{16x^2+81y^2}}$ ...(ii)

From (i) and (ii), we get $v_x=\frac{\mp 9yv}{\sqrt{16x^2+81y^2}}$

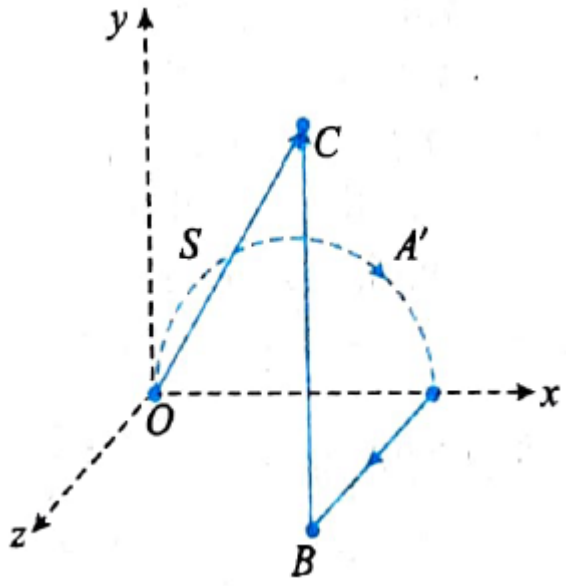
So, velocity is given by $\vec{v}=v_x\hat{i}+v_y\hat{j}=\frac{(\mp 9y\hat{i}\pm 4x\hat{j})v}{\sqrt{16x^2+81y^2}}$

Q.20 Answer:

$\frac{\pi+3}{3}$

Solution:

The distance D, that is, length of the actual path covered by the particle $(OA'ABC)$ as shown in figure.



$D=$ length of the semicircle $OA'A+$ length $AB+$ length BC. This gives $D=\pi R+R+2R=(\pi+3)R$

Since $OA=2R$, $AB=R$, and $BC=2R$, the coordinates of C can be given as $C\equiv(2R,R,2R)$.

Then the position of C is expressed as:

$\vec{r}_C=2R\hat{i}+2\hat{R}+R\hat{k}$

As $\vec{s}(=\overrightarrow{OC})=\vec{r}_C-\vec{r}_0$, substituting $\vec{r}_C$ and $\vec{r}_0=0\hat{i}+0\hat{j}+0\hat{k}$,

we obtain $\vec{s}=(2\hat{i}+2\hat{j}+\hat{k})R$.

Its magnitude $|\vec{s}|=\left(\sqrt{2^2+2^2+1^2}\right)R=3R$

Hence, $\frac{D}{s}=\frac{(\pi+3)R}{3R}=\frac{\pi+3}{3}$

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